Gait & Posture 62 (2018) 440–444

Contents lists available at ScienceDirect

Gait & Posture

journal homepage: www.elsevier.com/locate/gaitpost

Full length article

Patellar malalignment treated with modified knee extension training: An electromyography study

Chi-Chuan Wu[a,*], Mei-Chuan Chen[a], Po-Yuan Tseng[b], Chi-Heng Lu[b,c], Chiu-Ching Tuan[b]

[a] Department of Orthopedic Surgery, Chang Gung Memorial Hospital, Chang Gung University, Kweishan, Taoyuan, Taiwan
[b] Graduate Institute of Electronics Engineering, National Taipei University of Technology, Taipei, Taiwan
[c] Department of Radiation Oncology, Chang Gung Memorial Hospital, Kweishan, Taoyuan, Taiwan

ARTICLE INFO

Keywords:
Knee extension training
Patellar malalignment
Vastus medialis obliquus

ABSTRACT

Background: Patellar malalignment (PM) in most patients is ascribed to an imbalance of peripatellar soft tissue tension.
Research question: Conservative treatment of PM initially with enforced training of the vastus medialis obliquus (VMO) has been widely applied. Non-operative techniques for treatment of PM require continuing development.
Methods: Thirty healthy young adults participated in the study. Two surface electromyography (EMG) electrodes were placed on the skin of the dominant lower thigh in each subject: one on the center of the muscle belly of the VMO and the other on the symmetric location of the vastus lateralis (VL). Maximum of knee extension action (from various angles of knee flexion to full extension) was initiated. Tests were conducted with knee flexion decreasing by 10° at each step. Each action was repeated three times, and the average value was calculated. The root mean square value of excited muscles in the EMG was recorded. The ratio of the VMO to the VL (VMO/VL) was used to indicate the effectiveness of the treatment. The knee position varied from 90° flexion initially, decreasing by 10° at each step.
Results: Nine sets of values were obtained. All extension actions were effective (VMO/VL > 1; range, 1.23–1.35). The maximal value was observed at 60° flexion (VMO/VL = 1.35). Differences were not significant among the nine groups ($p = 0.08$, ANOVA).
Significance: Using the described knee extension training for conservative treatment of PM may be an effective alternative. The technique is simple, and the results of our experimental tests are encouraging. This method may become another popular and effective technique for treating PM.

1. Introduction

Patients with patellar malalignment (PM) are common in orthopedic clinics [1]. Although PM may occur in people of all ages, young female patients with PM are especially common [2]. PM usually presents with lateral patellar subluxation (LPS) [3]. The contributing factors of LPS have been determined: imbalance of peripatellar soft tissue tension, abnormality of bony structures, and abnormal skeletal axis [4,5]. Clinically, LPS in most patients is due to imbalanced peripatellar soft tissue tension, and the latter two causes are relatively rare [6].

Conservative treatment for PM due to imbalanced peripatellar soft tissue tension has been widely supported [7]. The reported success rate is as high as 87% [8]. Patellar bracing or strengthening training for the vastus medialis obliquus (VMO) is most effective [1,9]. However, methods for increasing the strength of the VMO are varied, and none has been shown to be clearly superior to the others [10]. In the literature, action of knee extension has been reported to be able to initiate muscle activity of the VMO in the final 0°–15° [11]. Furthermore, a goosestep gait is reported to be able to ameliorate PM [12]. A hypothesis is therefore created: some ranges of knee extension actions may be effective for treating PM. Additionally, surface electromyograms (EMG) have been successfully to detect muscle activity [13,14]. Using surface EMG, it may be possible to objectively validate whether peripatellar muscle activities are initiated individually. This prospective study had two aims: (1) to confirm whether knee extension action can enhance muscle strength in the medial aspect of the patella; (2) to find an optimal angle of knee position that may subsequently produce the maximal effectiveness of extension action for treating PM.

* Corresponding author at: Department of Orthopedic Surgery, Chang Gun Memorial Hospital, 5 Fu-Hsin St., Kweishan, Taoyuan, Taiwan.
E-mail address: ccwu@mail.cgu.edu.tw (C.-C. Wu).

https://doi.org/10.1016/j.gaitpost.2018.04.005
Received 6 March 2017; Received in revised form 29 March 2018; Accepted 5 April 2018
0966-6362/ © 2018 Published by Elsevier B.V.

2. Methods

2.1. Participants

From March 2016 to June 2016, 30 healthy adults (25 males and 5 females; average age 26.2 years; range 23–48 years) participated in this study. All subjects worked or studied at a single university. Subjects were enrolled due to the convenience of testing. None had any physical disorder. Using healthy adults for the study allowed complete focus on the strengthening of muscles without disturbance from associated pain. Patellar malalignment was diagnosed from clinical features. Images were often normal. In patients with patellofemoral pain, the etiology may not be completely due to patellar malalignment. The validity of the test thus might be compromised due to severe pain during forceful extension of the knee.

2.2. Methodology

2.2.1. EMG procedures

The SHIELD-EKG-EMG (Olimex Ltd., Plovdiv, Bulgaria) was used in this study. Before attachment of the surface EMG electrodes, the dominant lower thigh in each subject was swabbed with isopropyl alcohol [15]. The electrodes were placed on the center of the muscle belly of the VMO and the symmetric location of the vastus lateralis (VL) (Fig. 1).

The current device included three components: (1) an EMG sensor (SHIELD-EKG-EMG), (2) a microcontroller unit (MCU) (Arduino nano-328; HAOYU Electronics, Shenzhen, China), and (3) an angle sensor (Inertial Measure Unit (IMU), nine Degree-of-Freedom (9-DOF) GY-85; HAOYU Electronics, Shenzhen, China). Two sets of EMG systems were equipped, one for the VMO and the other for the VL. The SHIELD-EKG-EMG's total gain is the product of the gains at each discretization stage: an instrumental amplifier (G1 = 10), an Op Amp with regulated gain (G2 = 50), and a third-order "Besselworth" filter (G3 = 3.56) for which the cutoff frequency was set to fc = 40 Hz. The raw EMG signals are fed to the A/D conversion electronics linked to a computer with a sampling rate of 1024 Hz. The signals are rectified and bandpass filtered (20–400 Hz) to gain a linear envelope using the LabView (National Instruments, Austin, TX, USA) program.

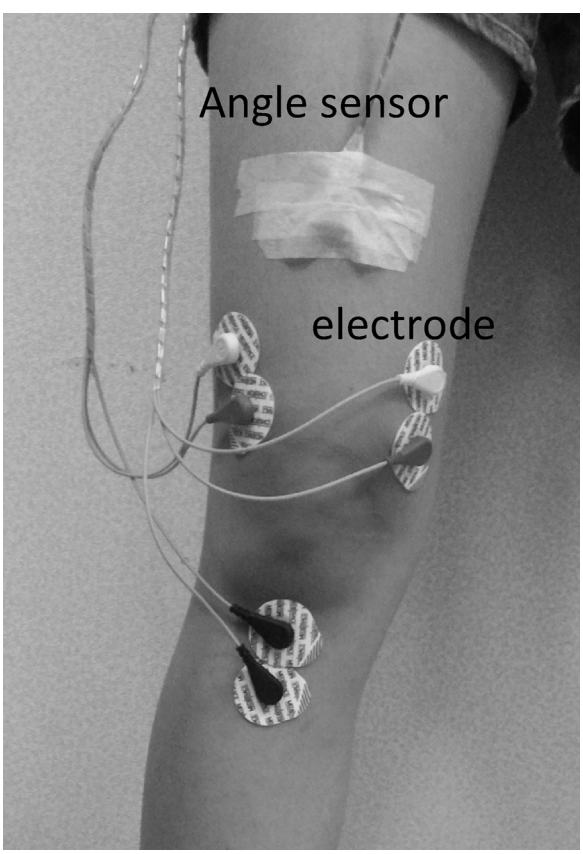

Fig. 1. The electrodes are placed at the center of the muscle belly of the VMO (vastus medialis obliquus) and the symmetric location of the vastus lateralis (VL). An angle sensor is placed at the anterior aspect of the lower thigh.

The data of knee extension were sequestered into segments. The root mean square (RMS) value of each segment was calculated. The intensity of the VMO and VL was presented as the VMO/VL ratio. A ratio greater than one indicated that the VMO was exerting a higher muscular contractile intensity than the VL, indicating that the training for enhancement of medial patellar structures was efficient.

2.2.2. Data collection

Each subject stood and held onto the wall with the contralateral hand. This action prevented extraneous movement of the trunk during the testing. The knee was lifted with hip flexion and ankle plantar flexion. First, the knee was positioned at 90° with the ankle at full flexion (Fig. 2). The knee was then fully extended, and the RMS value was recorded. Three extension actions were performed at each flexion angle of the knee, and the interval between each action was four seconds. The average of the three values was calculated and recorded. The tests were repeated with the knee positioned at each 10° decrease from 90° (i.e. 90°, 80°, …, 10°). The interval during each angular change was two minutes. Thus, nine sets of data were obtained, and each data set included 30 VMO/VL values. The entire testing course for the quadriceps femoris employed isotonic contraction. Concurrently, the knee angle was controlled by the angle sensor (Figs. 1 and 2) to maintain knee stability. The test positions of knee flexion were not randomly ordered; this was done in order to minimize errors, as frequent changes of the various flexion angles might have confused the examiners. The training with knee flexion from 90° to 10° was not geared to a specific purpose, but rather to simplify the procedure and minimize errors during the test.

2.3. Statistical analysis

Dependent samples for analysis of variance (ANOVA) were used for comparison among the nine sets of values using the SPSS statistical software (Version 17.0, SPSS Inc., Chicago, IL). P values less than 0.05 indicated statistical significance. If the comparison was significant, post hoc tests were used to further test for differences between sets.

3. Results

All 30 subjects finished the knee extension examinations, and nine sets of data were obtained (Table 1).

The VMO/VL ratios of the nine sets of data from 90° to 10° knee flexion were 1.29 (0.22), 1.30 (0.22), 1.33 (0.24), 1.35 (0.24), 1.34 (0.26), 1.27 (0.26), 1.23 (0.28), 1.25 (0.30), and 1.28 (0.33), respectively. The values indicated the mean (standard deviation) (Fig. 3).

Statistical comparison using dependent samples for ANOVA showed that the difference among ratios was marginally non-significant ($p = 0.08$). The maximum value of the mean VMO/VL ratio among the nine sets of data was 1.35, which was at 60° of knee flexion.

There were no thigh or knee complaints from any of the subjects throughout the testing course.

4. Discussion

The principle of using muscle strengthening to treat PM focuses on enhancing the strength of the VMO as far beyond the strength of the VL as possible [16]. The mechanisms of the VMO training in treating PM may include (1) the VMO muscle belly volume exceeding that of the VL

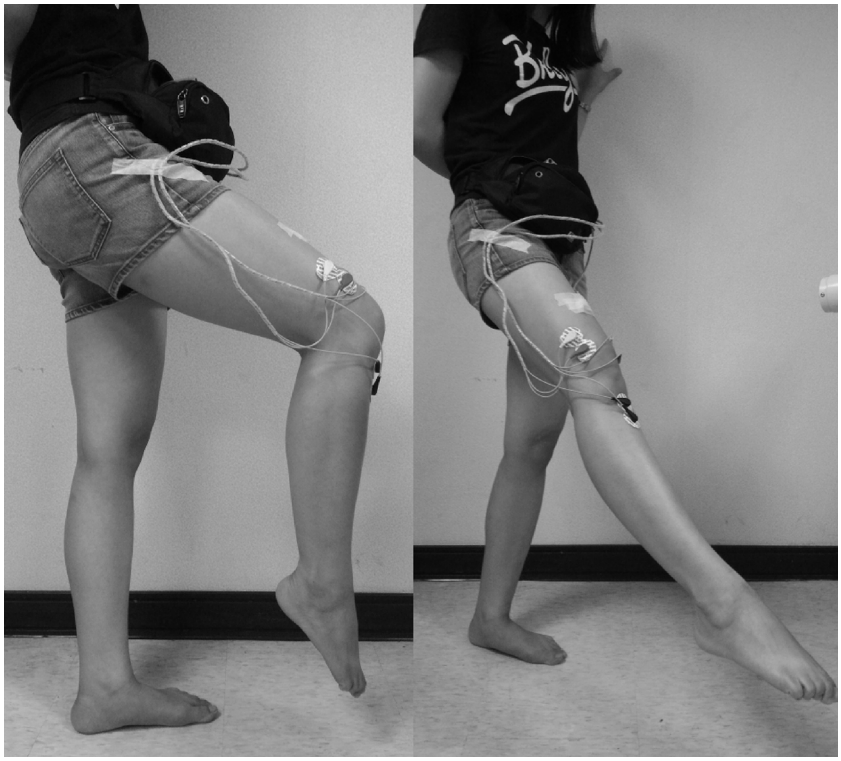

Fig. 2. Each subject stands and holds onto the wall with the contralateral hand. The hip and knee are upraised. The knee is positioned at a certain angle with the ankle at full flexion. The knee is then forcefully extended.

Table 1
VMO/VL ratio following various knee flexion angles to full extension of the knee.

	Knee flexion angle (degs)								
	10	20	30	40	50	60	70	80	90
VMO/VL									
Mean	1.28	1.25	1.23	1.27	1.34	1.35	1.33	1.30	1.29
S.D.	0.33	0.30	0.28	0.26	0.26	0.24	0.24	0.22	0.22

Note: VMO: vastus medialis obliquus; VL: vastus lateralis; S.D.: standard deviation.

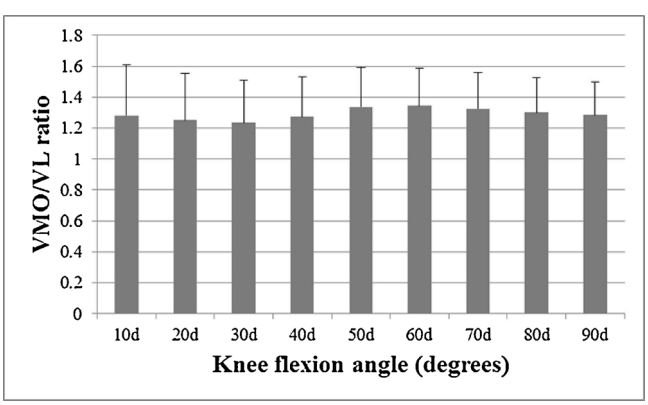

Fig. 3. Nine sets of data of VMO/VL ratio follow various knee flexion angles to full extension of the knee. Differences among angles were not significant (p = 0.08, ANOVA).

around the patella; (2) the pulling efficiency being greater in the VMO compared to the VL. Anatomically, the VMO normally has a relatively larger muscle belly volume and a greater intersecting angle (wider and more distal insertion to the patella) related to the quadriceps contractile axis compared to the VL (50° vs. 30°, Fig. 4) [11,17–19]. Therefore, pulling effects in the medial and lateral aspects of the patella are quite different. When the knee is extended, the VMO/VL ratio will exceed one, and the patella is pulled medially. The current study suggests that knee extension actions may introduce a 30% increment of inward pulling forces (a VMO/VL ratio of 1.23–1.35).

The current study did not determine an optimal knee position in which to start the knee extension action. Although all VMO/VL ratios were similar, clinical knee flexion of 50°–60° seems to be the most comfortable when the hip and knee are upraised [20,21]. Despite the result that the differences were not statistically significant, 50°–60° of knee flexion may achieve a maximal VMO/VL ratio (Table 1, Fig. 3). Why was the ankle recommended to be flexed downwards when the knee is extended in the current study? The authors had tested both positions with the ankle in flexion and dorsiflexion. Anatomically, the gastrocnemius connects both the knee and ankle joints in the posterior aspect of the leg [22]. If the ankle is dorsiflexed while the knee is forcefully extended, the gastrocnemius will be suddenly stretched, which may cause discomfort or even injury.

To perform the current technique, one hand holding a fixed object is strongly recommended. When the knee is forcefully extended, maintenance of body balance is important, and all forces can then be concentrated on the knee. Body imbalance may prevent the training or even causes a sports injury. The test is performed in the standing position in order to allow the quadriceps femoris be freely strengthened.

In 1968, Lieb and Perry found that the maximal VMO activity occurred during the final 15° of knee extension [11]. In 1995, Cerny used EMG with indwelling wire electrodes to determine which exercise might improve the VMO/VL ratio [23]. The latter study found that only knee extension training was effective. Moreover, the knee extension had to be combined with medial rotation and not lateral rotation of the hip. The maximal VMO/VL ratio in that study was 1.2. In addition, the effectiveness of walk stance-step down training was minimal. In 2009, Wu et al. suggested that goosestep training (a gait with forcefully full extension of the knee that is normally performed during formal military

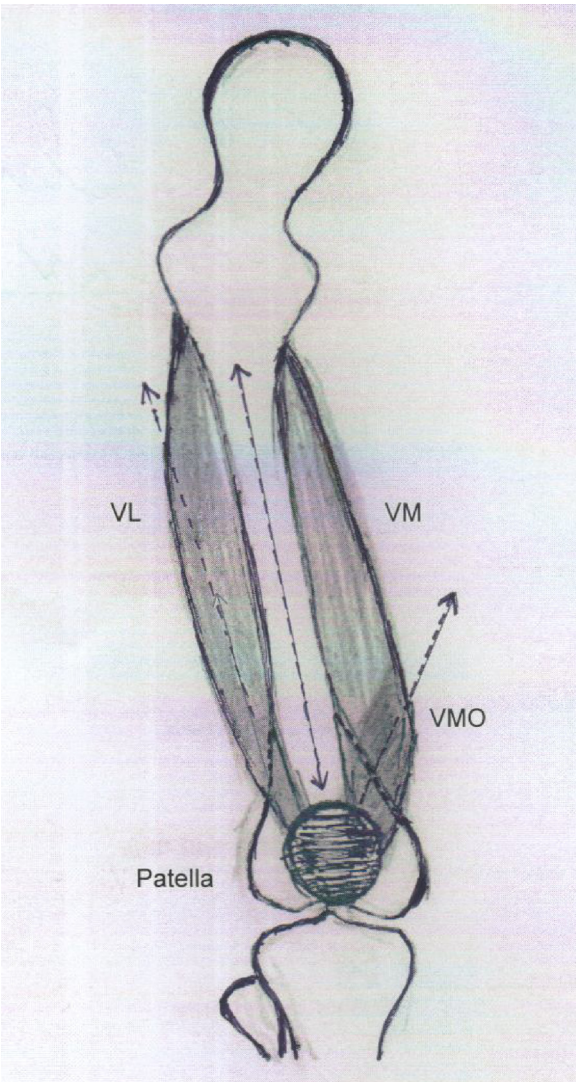

Fig. 4. xxx.

parades) improved 72.7% of patients with PM [12]. These prior studies may support the effectiveness of knee extension training. In the current study, forceful knee extension achieved VMO/VL ratios of 1.23–1.35.

In 2000, Sakai et al. in a cadaveric study suggested that a weak VMO could introduce PM [24]. There are numerous techniques for strengthening the VMO to treat PM. Using electrical stimulation to strengthen the quadriceps femoris, both Laughman et al. (1983) and Caggiano et al. achieved significant increases in muscle torque [25,26]. Because patients required treatment at a specially equipped location, the practicality of this technique is doubtful. In contrast, knee extension training recommended in the current study can be performed at any time and place.

In 1998, Sheehy et al. using a surface EMG study showed that patellofemoral pain was unrelated to the VMO/VL ratio [27]. In 2003, Mohr et al. using a fine wire EMG study suggested that patellofemoral pain was due to weakness of the entire quadriceps muscle group and not simply the VMO [28]. Because patellofemoral pain may be caused by a number of etiologies, simply strengthening the VMO activity may not alleviate patellofemoral pain [4,9,29,30]. However, in theory, if patellofemoral pain is caused by PM, enhancing the strength of the VMO should be effective [7,13,17]. If so, knee extension action should be an effective treatment. In any event, the current study requires clinical confirmation before the method can enjoy wide application.

Knee extension may be performed with a closed or open chain, and clinical presentations may be quite different [13,31]. A closed chain can involve several joints and muscles. Although it is favored in most physical rehabilitation programs, treatment of PM via strengthening the VMO need not involve several structures of the lower extremities [14,32,33]. In contrast, an open chain is suitable for single muscle group training, and the action can be speedily and forcefully implemented. It therefore may be more appropriate in VMO strengthening [31,34]. In 2010, Irish et al. in a surface EMG study advocated open chain knee extension exercise to produce greater VL activity [35]. However, despite the fact that the current study used open chain training, the VMO/VL ratio was approximately 1.3. Therefore, more studies may be needed to validate either finding.

The current study has some limitations: (1) The sample size was small, and physical variation in patients may thus have influenced the results. Statistically non-significant differences may therefore have been due to the sample size. Enlarging the same size may allow an optimal flexion angle to be identified. (2) To have patients be familiar with knee extension action, a certain level of pre-test training may be necessary. After repeated exercises, the VMO/VL ratio may change. Pre-training may suggest an optimal knee position for extension. In other words, repeated training and EMG testing may be necessary. (3) There may be other positions for subjects in performing knee extension that are more effective. This may require further study. The current study may have created a feasible route for future research. (4) The current study only suggests the applicability of knee extension training in the treatment of PM. The results should be confirmed by sonographic morphology of the VMO [17,36,37]. This would help to validate the effectiveness of the treatment.

In conclusion, using the described knee extension training for conservative treatment of PM may be an effective alternative. The knee position may be initiated at any angle less than 90° of flexion. The technique is relatively simple, and our results are encouraging. Our proposed method may become another popular and effective technique for treating PM.

Conflict of interest statement

No authors had any financial and personal relationships with other people or organizations that could inappropriately influence (bias) this work.

References

[1] T.Y. Lan, W.P. Lin, C.C. Jiang, H. Chiang, Immediate effect and predictors of effectiveness of taping for patellofemoral pain syndrome: a prospective cohort study, Am. J. Sports Med. 38 (2010) 1626–1630.
[2] W. Petersen, A. Ellermann, A. Gösele-Koppenburg, R. Best, I.V. Rembitzki, G.P. Brüggemann, et al., Patellofemoral pain syndrome, Knee Surg. Sports Traumatol. Arthrosc. 22 (2014) 2264–2274.
[3] J.S. Mulford, C.J. Wakeley, J.D. Eldridge, Assessment and management of chronic patellofemoral instability, J. Bone Joint Surg. 89B (2007) 709–716.
[4] R.P. Grelsamer, Patellar malalignment, J. Bone Joint Surg. 82A (2000) 1639–1650.
[5] S.J. Rhee, G. Pavlou, J. Oakley, D. Barlow, F. Haddad, Modern management of patellar instability, Int. Orthop. 36 (2012) 2447–2456.
[6] E. Arendt, Anatomy and malalignment of the patellofemoral joint: its relation to patellofemoral arthrosis, Clin. Orthop. Relat. Res. 436 (2005) 71–75.
[7] A.C. Colvin, R.V. West, Patellar instability, J. Bone Joint Surg. Am. 90 (2008) 2751–2762.
[8] W.R. Post, Patellofemoral pain: results of nonoperative treatment, Clin. Orthop. Relat. Res. 436 (2005) 55–59.
[9] J.P. Fulkerson, Diagnosis and treatment of patients with patellofemoral pain, Am. J. Sports Med. 30 (2002) 447–456.
[10] P. Miao, Y. Xu, C. Pan, H. Liu, C. Wang, Vastus medialis oblique and vastus lateralis activity during a double-leg semisquat with or without hip adduction in patients with patellofemoral pain syndrome, BMC Musculoskelet. Disord. 12 (2015) 289.
[11] F.J. Lieb, J. Perry, Quadriceps function: an anatomic and mechanical study using amputated limbs, J. Bone Joint Surg. 50A (1968) 1535–1548.
[12] C.C. Wu, S.W. Chou, W.H. Hong, Effectiveness of goosestep training or its modification on treating patellar malalignment syndrome: clinical, radiographic, and electromyographic studies, Arch. Orthop. Trauma Surg. 129 (2009) 333–341.
[13] S.F. Tang, C.K. Chen, R. Hsu, S.W. Chou, W.H. Hong, H.L. Lew, Vastus medialis obliquus and vastus lateralis activity in open and closed kinetic chain exercises in

patients with patellofemoral pain syndrome: an electromyographic study, Arch. Phys. Med. Rehabil. 82 (2001) 1441–1445.
[14] W.G. Yoo, Effects of the slow speed-targeting squat exercise on the vastus medialis oblique/vastus lateralis muscle ratio, J. Phys. Ther. Sci. 27 (2015) 2861–2862.
[15] J.V. Basmajian, R. Blumenstein, Electrode placement in EMG biofeedback, in: J.V. Basmajian (Ed.), Biofeedback: Principles and Practice for Clinicians, Williams & Wilkins, Baltimore, MD, 1989, pp. 363–377.
[16] J.E. Earl, A.Z. Hoch, A proximal strengthening program improves pain, function, and biomechanics in women with patellofemoral pain syndrome, Am. J. Sports Med. 39 (2011) 154–163.
[17] M.H. Jan, D.H. Lin, J.J. Lin, C.H. Lin, C.K. Cheng, Y.F. Lin, Differences in sonographic characteristics of the vastus medialis obliquus between patients with patellofemoral pain syndrome and healthy adults, Am. J. Sports Med. 37 (2009) 1743–1749.
[18] M. Schuenke, E. Schulte, U. Schumacher, Atlas of Anatomy: General Anatomy and Musculoskeletal System, Thieme Stuttgart, Germany, 2010.
[19] C.M. Powers, R.B. Souza, J.P. Fulkerson, Patellofemoral joint, in: D.T. Magee, J.E. Zachazewski, W.S. Quillen, R.C. Manske (Eds.), Pathology and Intervention in Musculoskeletal Rehabilitation, 2nd, Elsevier, Maryland Heights MO, 2015, pp. 798–835.
[20] F.H. Fu, M.G. Maday, Arthroscopic lateral release and the lateral patellar compression syndrome, Orthop. Clin. N. Am. 23 (1992) 601–612.
[21] E.A.P. de Moura Campos Carvalho Silva, E. Magalhães, F.F. Bryk, T.Y. Fukuda, Comparison of isometric ankle strength between females with and without patellofemoral pain syndrome, Int. J. Sports Phys. Ther. 9 (2014) 628–634.
[22] A.M.R. Agur, A.F. Dalley II, Grant's Atlas of Anatomy, 13th, Wolters Kluwer/Lippincott Williams & Wilkins, Philadelphia, PA, 2013.
[23] K. Cerny, Vastus medialis oblique/vastus lateralis muscle activity ratios for selected exercises in persons with and without patellofemoral pain syndrome, Phys. Ther. 75 (1995) 672–683.
[24] N. Sakai, Z.P. Luo, J.A. Rand, K.N. An, The influence of weakness in the vastus medialis oblique muscle on the patellofemoral joint: an in vitro biomechanical study, Clin. Biomech. 15 (2000) 335–339.
[25] R.K. Laughman, J.W. Youdas, T.R. Garrett, E.Y.S. Chao, Strength changes in the normal quadriceps femoris muscle as a result of electrical stimulation, Phys. Ther. 63 (1983) 494–499.
[26] E. Caggiano, T. Emrey, S. Shirley, R.L. Craik, Effects of electrical stimulation or voluntary contraction for strengthening the quadriceps femoris muscles in an aged male population, J. Orthop. Sports Phys. Ther. 20 (1994) 22–28.
[27] P. Sheehy, R.G. Burdett, J.J. Irrgang, J. VanSwearingen, An electromyographic study of vastus medialis oblique and vastus lateralis activity while ascending and descending steps, J. Orthop. Sports Phys. Ther. 27 (1998) 423–429.
[28] K.J. Mohr, R.S. Kvitne, M.M. Pink, B. Fideler, J. Perry, Electromyography of the quadriceps in patellofemoral pain with patellar subluxation, Clin. Orthop. Relat. Res. 415 (2003) 261–271.
[29] S.F. Dye, The pathophysiology of patellofemoral pain: a tissue homeostasis perspective, Clin. Orthop. Relat. Res. 436 (2005) 100–110.
[30] R.A. van der Heijden, N.E. Lankhorst, R. van Linschoten, S.M. Bierma-Zeinstra, M. van Middelkoop, Exercise for treating patellofemoral pain syndrome, Cochrane Database Syst. Rev. 1 (2015) CD010387.
[31] S.A. Doucette, D.D. Child, The effect of open and closed chain exercise and knee joint position on patellar tracking in lateral patellar compression syndrome, J. Orthop. Sports Phys. Ther. 23 (1996) 104–110.
[32] J.R. Blackburn, M.C. Morrissey, The relationship between open and closed kinetic chain strength of the lower limb and jumping performance, J. Orthop. Sports Phys. Ther. 27 (1998) 430–435.
[33] I. Cho, G. Hwangbo, D. Lee, S. Lee, The effect of closed kinetic chain exercises and open kinetic chain exercises using elastic bands on electromyographic activity in degenerative gonarthritis, J. Phys. Ther. Sci. 26 (2014) 1481–1484.
[34] G. Syme, P. Rowe, D. Martin, G. Daly, Disability in patients with chronic patellofemoral pain syndrome: a randomised controlled trial of VMO selective training versus general quadriceps strengthening, Man. Ther. 14 (2009) 252–263.
[35] S.E. Irish, A.J. Millward, J. Wride, B.M. Haas, G.L. Shum, The effect of closed-kinetic chain exercises and open-kinetic chain exercise on the muscle activity of vastus medialis oblique and vastus lateralis, J. Strength Cond. Res. 24 (2010) 1256–1262.
[36] Y.F. Lin, J.J. Lin, C.K. Cheng, D.H. Lin, M.H. Jan, Association between sonographic morphology of vastus medialis obliquus and patellar alignment in patients with patellar femoral pain syndrome, J. Orthop. Sports Phys. Ther. 38 (2008) 196–202.
[37] M. Khoshkhoo, A. Killingback, C.J. Robertson, P.J. Adda, The effect of exercise on vastus medialis oblique muscle architecture: an ultrasound investigation, Clin. Anat. 29 (2016) 752–758.